



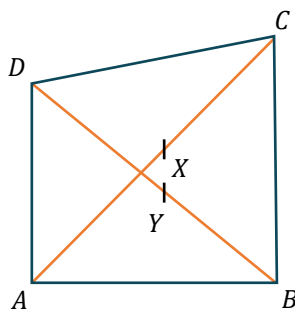
Vector : RACE-01

Single Correct Question

- The position vectors of two points A and B are $\hat{i} + \hat{j} - \hat{k}$ and $2\hat{i} - \hat{j} + \hat{k}$ respectively. Then $|\overrightarrow{AB}| =$
 (A) 2 (B) 3 (C) 4 (D) 5
MVT164
- The position vectors of P and Q are $5\hat{i} + 4\hat{j} + a\hat{k}$ and $-\hat{i} + 2\hat{j} - 2\hat{k}$ respectively. If the distance between them is 7, then the value of a will be
 (A) -5, 1 (B) 5, 1 (C) 0, 5 (D) 1, 0
MVT165
- If $\vec{a}$ and $\vec{b}$ are two vectors of magnitude 1 inclined at 120° , then the angle between $\vec{b}$ and $\vec{b} - \vec{a}$ is-
 (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{5\pi}{6}$
MVT166
- If $ABCD$ is a parallelogram, $\overrightarrow{AB} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\overrightarrow{AD} = \hat{i} + 2\hat{j} + 3\hat{k}$, then the unit vector in the direction of BD is
 (A) $\frac{1}{\sqrt{69}}(\hat{i} + 2\hat{j} - 8\hat{k})$ (B) $\frac{1}{69}(\hat{i} + 2\hat{j} - 8\hat{k})$ (C) $\frac{1}{\sqrt{69}}(-\hat{i} - 2\hat{j} + 8\hat{k})$ (D) $\frac{1}{69}(-\hat{i} - 2\hat{j} + 8\hat{k})$
MVT167

Subjective Type Question

- Let point A, B, C, D are vertices of parallelogram $ABCD$. Then prove that
 $2\overrightarrow{AB} + 3\overrightarrow{BC} + 4\overrightarrow{CD} + 5\overrightarrow{DA} = 2\overrightarrow{CA}$.
MVT168
- Let O be the centre of a regular pentagon $ABCDE$ and $\overrightarrow{OA} = \vec{a}$. Then prove that
 $\overrightarrow{AB} + 2\overrightarrow{BC} + 3\overrightarrow{CD} + 4\overrightarrow{DE} + 5\overrightarrow{EA}$ equals $5\vec{a}$.
MVT169
- In figure, X and Y are the mid-points of AC and BD . Write down an expression for $\overrightarrow{AY}$ in terms of $\overrightarrow{AD}$ and $\overrightarrow{AB}$. Write down an expression for $\overrightarrow{YX}$ in terms of $\overrightarrow{YA}$ & $\overrightarrow{YC}$. Show that
 $\overrightarrow{AB} + \overrightarrow{AD} + \overrightarrow{CB} + \overrightarrow{CD} = 4 \bullet \overrightarrow{XY}$.



MVT170

8. If D, E and F are the mid-points of the sides BC, CA and AB , respectively of $\triangle ABC$ and O is any point, then prove the following.

$$(a) \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} = \overrightarrow{OD} + \overrightarrow{OE} + \overrightarrow{OF}$$

$$(b) \overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF} = 0$$

$$(c) \overrightarrow{OE} + \overrightarrow{OF} + \overrightarrow{DO} = \overrightarrow{OA}$$

$$(d) \overrightarrow{AD} + \frac{2}{3}\overrightarrow{BE} + \frac{1}{3}\overrightarrow{CF} = \frac{1}{2}\overrightarrow{AC}$$

MVT171

9. If M and N are the mid points of the sides BC and CD respectively of a parallelogram $ABCD$, then $\overrightarrow{AM} + \overrightarrow{AN} = \lambda \overrightarrow{AC}$; where λ is a scalar. Find λ .

MVT172

10. If $\vec{a} = 2\hat{i} - 2\sin\theta\hat{j} + 3\hat{k}$ and $\vec{b} = 4\cos\alpha\hat{i} - \hat{j} + 2\sqrt{3}\cos\theta\hat{k}$ are equal vectors then find general values of ' θ ' & ' α '.

MVT173

11. Find all values of λ such that $(x, y, z) \neq (0, 0, 0)$ and $(\vec{i} + \vec{j} + 3\vec{k})x + (3\vec{i} - 3\vec{j} + \vec{k})y + (-4\vec{i} + 5\vec{j})z = \lambda(\vec{i}x + \vec{j}y + \vec{k}z)$, where $\vec{i}, \vec{j}, \vec{k}$ are unit vectors along the coordinate axes.

MVT174

ANSWER KEY

Question	1	2	3	4	9	10	11
Answer	B	A	C	C	$\frac{3}{2}$	$\theta = 2n\pi + \frac{\pi}{6}, \alpha = 2n\pi \pm \frac{\pi}{3}$	$(0, -1)$



Vector : RACE-02

Multiple Correct Question

1. A line passes through a point A with position vector $3\hat{i} + \hat{j} - \hat{k}$ and is parallel to the vector $2\hat{i} - \hat{j} + 2\hat{k}$. If P is a point on this line such that $AP = 15$ units, then the position vector of the point P is/are
- (A) $13\hat{i} + 4\hat{j} - 9\hat{k}$ (B) $13\hat{i} - 4\hat{j} + 9\hat{k}$ (C) $7\hat{i} - 6\hat{j} + 11\hat{k}$ (D) $-7\hat{i} + 6\hat{j} - 11\hat{k}$

MVT175

Subjective Type Question

2. Find the position vectors of the points which divide the join of the points $2\vec{a} - 3\vec{b}$ and $3\vec{a} - 2\vec{b}$ internally and externally in the ratio $2 : 3$.

MVT176

3. P, Q have position vectors $\vec{a}$ and $\vec{b}$ relative to the origin 'O' and X, Y divide $\overline{PQ}$ internally and externally respectively in the ratio $2 : 1$. Find vector $\overline{XY}$ in terms of $\vec{a}, \vec{b}$ -

MVT177

4. Let $\vec{p}$ is the p.v. of the orthocentre & $\vec{g}$ is the p.v. of the centroid of the triangle ABC where circumcentre is at origin. If $\vec{p} = K\vec{g}$, then find value of K .

MVT178

5. Let S, G and H be the circumcenter, centroid and orthocenter respectively of triangle ABC . Then prove the following :

- (i) $\overrightarrow{PA} + \overrightarrow{PB} + \overrightarrow{PC} = 3\overrightarrow{PG}$, where P is any point in the plane of triangle ABC .
 (ii) $\overrightarrow{SA} + \overrightarrow{SB} + \overrightarrow{SC} = \overrightarrow{SH}$.
 (iii) $\overrightarrow{HA} + \overrightarrow{HB} + \overrightarrow{HC} = 2\overrightarrow{HS}$.

MVT179

6. If the vectors $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = -\hat{i} + 3\hat{j} + 4\hat{k}$ & $\vec{c} = 4\hat{i} - 2\hat{j} - 6\hat{k}$ constitute the sides $\overline{CB}, \overline{CA}$ & $\overline{AB}$ of a ΔABC , then find

- (a) The length of the median bisecting the vector $\vec{c}$.
 (b) Length of internal angle bisector of angle A .

MVT180

7. The position vectors of the points A, B, C where ABC is a triangle are $\vec{a}, \vec{b}$ and $\vec{c}$ respectively. E is a point in AB such that $AE : EB = 2 : 3$. Find the position vector of the point P where CE meets the median AD in terms of the vectors $\vec{a}, \vec{b}$ and $\vec{c}$.

MVT181

Vector

8. Find the vector equation of the line which is parallel to the vector $3\hat{i} - 2\hat{j} + 6\hat{k}$ and which passes through the point $(1, -2, 3)$. MVT182
9. Find the unit vectors along the angle bisector between the vectors $\vec{a} = \hat{i} + 2\hat{j} - 2\hat{k}$ and $\vec{b} = -3\hat{i} + 6\hat{j} + 2\hat{k}$. MVT183
10. Find point of intersection of lines $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 4\hat{j} + 5\hat{k})$ and $\vec{r} = 3\hat{i} + 6\hat{k} + 8\hat{k} + \mu(6\hat{i} - 7\hat{j} + 8\hat{k})$ MVT184
11. Find point of intersection of line $\vec{r} = 2\hat{i} - \hat{j} - 3\hat{k} + \lambda(-\hat{i} + 2\hat{j} + 2\hat{k})$ and plane $\vec{r} = (5\hat{i} - 7\hat{j} - 9\hat{k}) + \mu(\hat{i} - 2\hat{j}) + \nu(3\hat{k} - \hat{j})$ MVT185
12. Let $\vec{r} = \vec{a} + \lambda\vec{l}$ and $\vec{r} = \vec{b} + \mu\vec{m}$ be two lines in space where $\vec{a} = 5\hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = -\hat{i} + 7\hat{j} + 8\hat{k}$, $\vec{l} = -4\hat{i} + \hat{j} - \hat{k}$ and $\vec{m} = 2\hat{i} - 5\hat{j} - 7\hat{k}$ then find the p.v. of a point which lies on both of these lines. MVT186

ANSWER KEY

- | | | | | | | | |
|-----|--|-----|--|-----|---|----|---|
| 1. | B,D | 2. | $(-5\vec{b})$ | 3. | $\frac{4\vec{b} - 4\vec{a}}{3}$ | 4. | 3 |
| 6. | (a) $\sqrt{6}$ (b) $\frac{3}{4}\sqrt{10}$ | 7. | $\frac{3\vec{a}}{7} + \frac{2\vec{b}}{7} + \frac{2\vec{c}}{7}$ | 8. | $\vec{r} = (\hat{i} - 2\hat{j} + 3\hat{k}) + \lambda(3\hat{i} - 2\hat{j} + 6\hat{k})$ | | |
| 9. | $\frac{-\hat{i} + 16\hat{j} - 4\hat{k}}{\sqrt{273}}$ | 10. | (3, 6, 8) | 11. | (5, -7, -9) | | |
| 12. | (1, 2, 1) | | | | | | |



Vector : RACE-03

Subjective Type Question

1. Find the angle between two vectors $\vec{a}$ & $\vec{b}$ with magnitude 2 and 1 respectively and such that $\vec{a} \cdot \vec{b} = \sqrt{3}$.
MVT187
2. If $\vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j}$ are such that $\vec{a} + \lambda\vec{b}$ perpendicular to $\vec{c}$ then the value of λ is
MVT188
3. The scalar product of the vector $\hat{i} + \hat{j} + \hat{k}$ with a unit vector along the sum of the vectors $2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\lambda\hat{i} + 2\hat{j} + 3\hat{k}$ is equal to 1, find λ .
MVT189
4. If the vectors $3\vec{p} + \vec{q}$; $5\vec{p} - 3\vec{q}$ and $2\vec{p} + \vec{q}$; $4\vec{p} - 2\vec{q}$ are pairs of mutually perpendicular vectors then find $\sin(\vec{p} \wedge \vec{q})$.
MVT190
5. Find the acute angle between the medians drawn from the acute angles of an isosceles right angled triangle.
MVT191
6. Find the set of values of c for which the angle between the vectors $c x \hat{i} - 6 \hat{j} + 3 \hat{k}$ & $x \hat{i} - 2 \hat{j} + 2 c x \hat{k}$ is acute for every $x \in R$.
MVT192
7. Let $\vec{a}, \vec{b}, \vec{c}$ be vectors of length 3, 4, 5 respectively. Let $\vec{a}$ be perpendicular to $\vec{b} + \vec{c}$, $\vec{b}$ to $\vec{c} + \vec{a}$ & $\vec{c}$ to $\vec{a} + \vec{b}$. Then find $|\vec{a} + \vec{b} + \vec{c}|$.
MVT193
8. Let $\vec{u} = \hat{i} + \hat{j}$, $\vec{v} = \hat{i} - \hat{j}$ and $\vec{w} = \hat{i} + 2\hat{j} + 3\hat{k}$ If $\hat{n}$ is a unit vector such that $\vec{u} \cdot \hat{n} = 0$ and $\vec{v} \cdot \hat{n} = 0$, then find $|\vec{w} \cdot \hat{n}|$.
MVT194
9. Find the projection of the vector $\vec{a} = 4\hat{i} - 2\hat{j} + \hat{k}$ on the vector $\vec{b} = 3\hat{i} + 6\hat{j} + 2\hat{k}$. Also find vector component of along $\vec{a}$ and $\vec{b}$ perpendicular to $\vec{b}$.
MVT195
10. Prove that in a regular tetrahedron any pair of opposite edges are perpendicular.
MVT196
11. If $2\sin x \sin y + 3 \cos y + 6 \cos x \sin y = 7$, then find the value of $\tan^2 x + 2 \tan^2 y$.
MVT197

12. Given a parallelogram $ABCD$. If $|\vec{AB}| = a$, $|\vec{AD}| = b$ & $|\vec{AC}| = c$, then find $\vec{DB} \cdot \vec{AB}$. MVT198
13. In the isosceles triangle ABC $|\vec{AB}| = |\vec{BC}| = 8$, a point E divides AB internally in the ratio $1 : 3$, then find the cosine of the angle between $\vec{CE}$ & $\vec{CA}$ (where $|\vec{CA}| = 12$) MVT199
14. L_1 and L_2 are two lines whose vector equations are
 $L_1 : \vec{r} = \lambda \left[(\cos \theta + \sqrt{3})\hat{i} + (\sqrt{2} \sin \theta)\hat{j} + (\cos \theta - \sqrt{3})\hat{k} \right]$
 $L_2 : \vec{r} = \mu (a\hat{i} + b\hat{j} + c\hat{k}),$
 where λ and μ are scalars and α is the angle between L_1 and L_2 . If the angle ' α ' is independent of θ then find the value of ' α ' MVT200

ANSWER KEY

- | | | | |
|---|--|----------------------------------|--------------------------|
| 1. $\frac{\pi}{6}$ | 2. 8 | 3. 1 | 4. $\frac{\sqrt{55}}{8}$ |
| 5. $\cos^{-1} \frac{4}{5}$ | 6. $c \in \left[0, \frac{4}{3} \right)$ | 7. $5\sqrt{2}$ | 8. 3 |
| 9. $\frac{190\hat{i} - 110\hat{j} + 45\hat{k}}{49}$ | 11. 9 | 12. $\frac{3a^2 - c^2 + b^2}{2}$ | |
| 13. $\frac{3\sqrt{7}}{8}$ | 14. 30° | | |



Vector : RACE-04

Single Correct Question

- If $\vec{a} = (1, -1, 1)$ and $\vec{c} = (-1, -1, 0)$, then the vector $\vec{b}$ satisfying $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{a} \cdot \vec{b} = 1$ is
(A) $(1, 0, 0)$ (B) $(0, 0, 1)$ (C) $(0, -1, 0)$ (D) None of these
MVT201
- For two particular vectors $\vec{A}$ and $\vec{B}$ it is known that $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$. What must be true about the two vectors?
(A) At least one of the two vectors must be the zero vector.
(B) $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$ is true for any two vectors.
(C) One of the two vectors is a scalar multiple of the other vector.
(D) The two vectors must be perpendicular to each other.
MVT202
- Find the number of vectors of unit length perpendicular to vectors $\vec{a} = (1, 1, 0)$ & $\vec{b} = (0, 1, 1)$.
(A) 2 (B) 3 (C) 4 (D) ∞
MVT203

Multiple Correct Question

- Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $\vec{a} \cdot \vec{a} = \vec{b} \cdot \vec{b} = \vec{c} \cdot \vec{c} = 3$ and $|\vec{a} + \vec{b} - \vec{c}|^2 + |\vec{b} + \vec{c} - \vec{a}|^2 + |\vec{c} + \vec{a} - \vec{b}|^2 = 36$, then
(A) $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = \frac{-9}{2}$ (B) $\vec{a}, \vec{b}, \vec{c}$ are coplanar vectors
(C) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq \vec{0}$ (D) $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = \frac{-27}{2}$
MVT204

Subjective Type Question

- If $\vec{a}$ and $\vec{b}$ are vectors such that $|\vec{a} + \vec{b}| = \sqrt{29}$ and $\vec{a} \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = (2\hat{i} + 3\hat{j} + 4\hat{k}) \times \vec{b}$, value of $|(\vec{a} + \vec{b}) \cdot (-7\hat{i} + 2\hat{j} + 3\hat{k})|$ is
MVT205
- Find the area of the parallelogram whose diagonals are represented by the vectors $2\hat{i} + 3\hat{j} - 6\hat{k}$ and $3\hat{i} - 4\hat{j} - \hat{k}$.
MVT206
- If $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$, then show that $(\vec{a} - \vec{d})$ is parallel to $(\vec{b} - \vec{c})$ when $\vec{a} \neq \vec{d}$ and $\vec{b} \neq \vec{c}$.
MVT207

8. For any two vectors, $\vec{u}$ & $\vec{v}$ prove that

$$(a) (\vec{u} \cdot \vec{v})^2 + |\vec{u} \times \vec{v}|^2 = |\vec{u}|^2 |\vec{v}|^2 \quad (b) (1 + |\vec{u}|^2)(1 + |\vec{v}|^2) = (1 - \vec{u} \cdot \vec{v})^2 + |\vec{u} + \vec{v} + (\vec{u} \times \vec{v})|^2$$

MVT208

9. If $\vec{A}, \vec{B}, \vec{C}$ are unit vectors. Suppose that $\vec{A} \cdot \vec{B} = \vec{A} \cdot \vec{C} = 0$ and the angle between $\vec{B}$ and $\vec{C}$ is $\frac{\pi}{6}$. Prove that $\vec{A} = \pm 2(\vec{B} \times \vec{C})$.

MVT209

10. If A, B, C, D are any four points in space, prove that $|\vec{AB} \times \vec{CD} + \vec{BC} \times \vec{AD} + \vec{CA} \times \vec{BD}| = 4$ (Area of the triangle ABC)

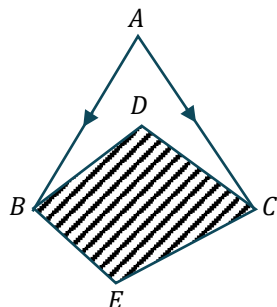
MVT210

11. $A_1, A_2, \dots, A_n$ are the vertices of a regular plane polygon with n sides and O is its centre.

$$\text{Show that } \sum_{i=1}^{n-1} (\vec{OA}_i \times \vec{OA}_{i+1}) = (1-n)(\vec{OA}_2 \times \vec{OA}_1)$$

MVT211

12. Let $\vec{AB} = 3\hat{i} - \hat{j}$, $\vec{AC} = 2\hat{i} + 3\hat{j}$ and $\vec{DE} = 4\hat{i} - 2\hat{j}$. Then, find the area of the shaded region in the figure



MVT212

13. Position vectors of the four angular points of a tetrahedron $ABCD$ are $A(3, -2, 1)$; $B(3, 1, 5)$; $C(4, 0, 3)$ and $D(1, 0, 0)$. Find the acute angle between the plane faces ADC and ABC .

MVT213

ANSWER KEY

Question	1	2	3	4	5	6	12	13	
Answer	B	C	A	ABC	4	$\frac{7\sqrt{26}}{2}$	7	$\theta = \tan^{-1} \frac{5}{2}$	



Vector : RACE-05

Single Correct Question

1. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ be three vectors. A vector $\vec{v}$ in the plane of $\vec{a}$ and $\vec{b}$, whose projection on $\vec{c}$ is $\frac{1}{\sqrt{3}}$, is given by
- (A) $\hat{i} - 3\hat{j} + 3\hat{k}$ (B) $-3\hat{i} - 3\hat{j} - \hat{k}$ (C) $3\hat{i} - \hat{j} + 3\hat{k}$ (D) $\hat{i} + 3\hat{j} - 3\hat{k}$

MVT214

Subjective Type Question

2. If $\vec{u}, \vec{v}, \vec{w}$ are non-coplanar vectors and p, q are integers, then find the number of ordered pairs (p, q) for which the equality $[3\vec{u} \ p\vec{v} \ p\vec{w}] - [p\vec{v} \ \vec{w} \ q\vec{u}] - [2\vec{w} \ q\vec{v} \ q\vec{u}] = 0$ holds.
- MVT215
3. If $\vec{r}$ be a vector perpendicular to $\vec{a} + \vec{b} + \vec{c}$, where $[\vec{a} \ \vec{b} \ \vec{c}] \neq 0$ and $\vec{r} = \ell(\vec{b} \times \vec{c}) + m(\vec{c} \times \vec{a}) + n(\vec{a} \times \vec{b})$, then find $\ell + m + n$.
- MVT216
4. If $\vec{a}, \vec{b}$ and $\vec{c}$ are non zero, non coplanar vectors determine whether the vectors $\vec{r}_1 = 2\vec{a} - 3\vec{b} + \vec{c}$, $\vec{r}_2 = 3\vec{a} - 5\vec{b} + 2\vec{c}$ and $\vec{r}_3 = 4\vec{a} - 5\vec{b} + \vec{c}$ are linearly independent or dependent.
- MVT217
5. Find a vector $\vec{r}$ in the plane of $\vec{p} = -\hat{i} + \hat{j}$ and $\vec{q} = -\hat{j} + \hat{k}$ such that $\vec{r}$ is perpendicular to $\vec{p}$ and $\vec{r} \cdot \vec{q} = -2$.
- MVT218
6. Let $\vec{a} = -\hat{i} - \hat{k}$, $\vec{b} = -\hat{i} + \hat{j}$ and $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then the value of $\vec{r} \cdot \vec{b}$ is
- MVT219
7. Suppose $\vec{v}_1, \vec{v}_2, \vec{v}_3$ is an orthogonal set of vectors in $\mathbb{R}^3$ with $\vec{v}_1 \cdot \vec{v}_1 = 29$, $\vec{v}_2 \cdot \vec{v}_2 = 85.25$, $\vec{v}_3 \cdot \vec{v}_3 = 9$, $\vec{w}$ is another vector in same space such that $\vec{w} \cdot \vec{v}_1 = -29$, $\vec{w} \cdot \vec{v}_2 = 341$, $\vec{w} \cdot \vec{v}_3 = -18$, then find $\vec{w}$ in terms of $\vec{v}_1, \vec{v}_2, \vec{v}_3$.
- MVT220
8. Find volume of parallelopiped formed by vectors $2\hat{i} - 3\hat{j} + 4\hat{k}$, $3\hat{j} + 2\hat{j} - 4\hat{k}$, $4\hat{i} + 3\hat{j} - 2\hat{k}$;
- MVT221
9. Find the volume of tetrahedron $ABCD$ with edge lengths $AB = 2$, $AC = 3$, $AD = 4$, $BC = \sqrt{13}$, $BD = 2\sqrt{5}$ and $CD = 5$.
- MVT222

10. Let M be the midpoint of AB in regular tetrahedron $ABCD$. What is $\cos(\angle CMD)$? MVT223
11. Find the shortest distance between the lines :
 $\vec{r}_1 = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$ & $\vec{r}_2 = (2\hat{i} + 4\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 4\hat{j} + 5\hat{k})$.- MVT224
12. Find the value of a and point of intersection of lines
 $\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = 2\hat{i} - \hat{j} - \hat{k} + \mu(2\hat{i} + \hat{j} + a\hat{k})$, given that they intersect. MVT225
13. A line passes through the point $A(\hat{i} + 2\hat{j} + 3\hat{k})$ and is parallel to the vector $\vec{V}(\hat{i} + \hat{j} + \hat{k})$. Then find its shortest distance from the origin. MVT226

ANSWER KEY

- | | | | | |
|-------------------|---|--|-----------------------|--|
| 1. C | 2. 1 | 3. 0 | 4. linearly dependent | 5. $\frac{2}{3}(\hat{i} + \hat{j} - 2\hat{k})$ |
| 6. 9 | 7. $-\vec{v}_1 + 4\vec{v}_2 - 2\vec{v}_3$ | 8. 50 | 9. 4 | |
| 10. $\frac{1}{3}$ | 11. $\frac{1}{\sqrt{6}}$ | 12. $a = \frac{13}{2}$ and $\left(\frac{10}{3}, \frac{-1}{3}, \frac{10}{3}\right)$ | 13. $\sqrt{2}$ | |



Vector : RACE-06

Subjective Type Question

- Find the volume of the parallelepiped whose coterminous edges are represented by $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = 3\hat{i} - \hat{j} + 2\hat{k}$.
MVT227
- The volume of the tetrahedron formed by the coterminous edges $\vec{a}$, $\vec{b}$, $\vec{c}$ is 3. Then find the volume of the parallelepiped formed by the coterminous edges $\vec{a} + \vec{b}$, $\vec{b} + \vec{c}$, $\vec{c} + \vec{a}$.
MVT228
- If $\vec{a} = 2\hat{i} - 4\hat{j} + 7\hat{k}$, $\vec{b} = 3\hat{i} + 5\hat{j} - 9\hat{k}$ and $\vec{c} = \hat{i} + \hat{j} + \hat{k}$, then find $[\vec{a} \vec{b} \vec{c}]$ and also $\vec{a} \times (\vec{b} \times \vec{c})$.
MVT229
- $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = 0$, then find the acute angle between $\vec{a}$ & $\vec{c}$.
MVT230
- Let $\vec{\alpha} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{\beta} = \hat{i} + \hat{j}$. If $\vec{\gamma}$ is a unit vector, then find the maximum value of $[\vec{\alpha} \times \vec{\beta} \vec{\beta} \times \vec{\gamma} \vec{\gamma} \times \vec{\alpha}]$.
MVT231
- If $\vec{a} = \frac{1}{\sqrt{10}}(3\hat{i} + \hat{k})$ and $\vec{b} = \frac{1}{7}(2\hat{i} + 3\hat{j} - 6\hat{k})$, then find the value of $(2\vec{a} - \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} + 2\vec{b})]$.
MVT232
- Given the following information about the non-zero vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$:
(i) $(\vec{A} \times \vec{B}) \times \vec{A} = \vec{0}$ (ii) $\vec{B} \cdot \vec{B} = 4$ (iii) $\vec{A} \cdot \vec{B} = -6$ (iv) $\vec{B} \cdot \vec{C} = 6$
Then find the value of $|\vec{A} \times \vec{B}| + |\vec{A}(\vec{B} \times \vec{C})| + |\vec{A} \cdot \vec{C}|$.
MVT233
- Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-zero unit vectors such that they are not-orthogonal pairwise. If $\vec{V}_1 = \vec{a} \times (\vec{b} \times \vec{c})$ and $\vec{V}_2 = (\vec{a} \times \vec{b}) \times \vec{c}$ are such that $\vec{V}_1 = \vec{V}_2$, then find $|(\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a})| + |(\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b})| + |(\vec{c} \times \vec{a}) \times (\vec{b} \times \vec{c})|$.
MVT234
- If $\vec{a} \cdot \vec{b} = 0$, $|\vec{a}| = 1$ and $|\vec{b}| = \frac{1}{2}$, then find $|\vec{a} \times (\vec{a} \times (\vec{a} \times (\vec{a} \times \vec{b})))|$.
MVT235

10. If $\vec{A} = 2\hat{i} + 3\hat{k}$, $\vec{B} = -2\hat{i} + \hat{j} + \hat{k}$, $\vec{C} = 10\hat{j} - \hat{k}$ determine x, y, z so that $(\vec{A} \times \vec{B}) \times \vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$
MVT236
11. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors, with $\vec{a} \wedge \vec{b} = \frac{\pi}{3} = \vec{b} \wedge \vec{c} = \vec{c} \wedge \vec{a}$ then find $[\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}]$.
MVT237
12. Vector $\vec{x}, \vec{y}, \vec{z}$ each of magnitude $\sqrt{2}$, make angles of 60° with each other. If $\vec{x} \times (\vec{y} \times \vec{z}) = \vec{a}$, $\vec{y} \times (\vec{z} \times \vec{x}) = \vec{b}$ and $\vec{x} \times \vec{y} = \vec{c}$ then find $\vec{x}, \vec{y}, \vec{z}$ in terms of $\vec{a}, \vec{b}, \vec{c}$.
MVT238

ANSWER KEY

1. 7 2. 36 3. $62, 92\hat{i} + 102\hat{j} + 32\hat{k}$ 4. $\frac{\pi}{6}$ 5. 3 6. -5
7. 9 8. 0 9. $\frac{1}{2}$ 10. $x = -9, y = -3, z = 0$
11. $\frac{1}{2}$ 12. $x = \frac{(\vec{a} + \vec{b}) \times \vec{c} + \vec{a} - \vec{b}}{-2}; y = \frac{(\vec{a} + \vec{b}) \times \vec{c} + \vec{a} + \vec{b}}{-2}; \vec{z} = \frac{(\vec{a} + \vec{b}) \times \vec{c} + \vec{a} - \vec{b}}{-2}$



Vector : RACE-07

Single Correct Question

1. Let $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ be two vectors. If $\vec{c}$ is a vector such that $\vec{b} \times \vec{c} = \vec{b} \times \vec{a}$ and $\vec{c} \cdot \vec{a} = 0$, then $\vec{c} \cdot \vec{b}$ is equal to

- (A) $\frac{1}{2}$ (B) -1 (C) $-\frac{1}{2}$ (D) $-\frac{3}{2}$

MVT239

2. Let $\vec{a} = \hat{i} - \hat{j}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{c} + \vec{b} = \vec{0}$ and $\vec{a} \cdot \vec{c} = 4$, then $|\vec{c}|^2$ is equal to

- (A) $\frac{19}{2}$ (B) 8 (C) $\frac{17}{2}$ (D) 9

MVT240

Subjective Type Question

3. If $\vec{p} = \frac{\vec{b} \times \vec{c}}{[\vec{b} \ \vec{c} \ \vec{a}]}$, $\vec{q} = \frac{\vec{c} \times \vec{a}}{[\vec{c} \ \vec{a} \ \vec{b}]}$, $\vec{r} = \frac{\vec{a} \times \vec{b}}{[\vec{a} \ \vec{b} \ \vec{c}]}$, then find the value of $(\vec{a} + \vec{b}) \cdot \vec{p} + (\vec{b} + \vec{c}) \cdot \vec{q} + (\vec{c} + \vec{a}) \cdot \vec{r}$.

MVT241

4. If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors and $\vec{a}' = \frac{\vec{b} \times \vec{c}}{[\vec{a} \ \vec{b} \ \vec{c}]}$, $\vec{b}' = \frac{\vec{c} \times \vec{a}}{[\vec{a} \ \vec{b} \ \vec{c}]}$ and $\vec{c}' = \frac{\vec{a} \times \vec{b}}{[\vec{a} \ \vec{b} \ \vec{c}]}$;

then prove that

- (a) $\vec{a}', \vec{b}', \vec{c}'$ are also not coplanar. (b) $(\vec{a}' \times \vec{b}') + (\vec{b}' \times \vec{c}') + (\vec{c}' \times \vec{a}') = \frac{\vec{a} + \vec{b} + \vec{c}}{[\vec{a} \ \vec{b} \ \vec{c}]}$.

MVT242

5. For any vector $\vec{r}$, prove that $\vec{r} = (\vec{r} \cdot \vec{a})\vec{a}' + (\vec{r} \cdot \vec{b})\vec{b}' + (\vec{r} \cdot \vec{c})\vec{c}'$ and $\vec{r} = (\vec{r} \cdot \vec{a}')\vec{a} + (\vec{r} \cdot \vec{b}')\vec{b} + (\vec{r} \cdot \vec{c}')\vec{c}$.

MVT243

6. If $\hat{a}, \hat{b}, \hat{c}$ are three non-coplanar unit vectors and α, β, γ are the angles between $\hat{a}$ and $\hat{b}$, $\hat{b}$ and $\hat{c}$, $\hat{c}$ and $\hat{a}$ respectively and $\vec{x}, \vec{y}, \vec{z}$ are unit vectors along the bisectors of the angle α, β, γ respectively,

then prove that $[\vec{x} \times \vec{y} \ \vec{y} \times \vec{z} \ \vec{z} \times \vec{x}] = \frac{1}{16} [a \ b \ c]^2 \sec^2 \frac{\alpha}{2} \sec^2 \frac{\beta}{2} \sec^2 \frac{\gamma}{2}$.

MVT244

7. If $\vec{d} = \lambda(\vec{a} \times \vec{b}) + \mu(\vec{b} \times \vec{c}) + \nu(\vec{c} \times \vec{a})$, $[\vec{a} \ \vec{b} \ \vec{c}] = \frac{1}{8}$ and $\vec{d} \cdot (\vec{a} + \vec{b} + \vec{c}) = 8$, then find $\lambda + \mu + \nu$.

MVT245

8. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ & $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$, then find the vector $\vec{c}$ such that $\vec{a} \cdot \vec{c} = 0$ & $\vec{a} \times \vec{c} = \vec{b}$.

MVT246

9. Let $\vec{a} = \hat{j} - \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$. Then find the vector $\vec{b}$ satisfying $\vec{a} \times \vec{b} + \vec{c} = \vec{0}$ and $\vec{a} \cdot \vec{b} = 3$.
MVT247
10. The vectors $\vec{a}$ and $\vec{b}$ are not perpendicular and $\vec{c}$ and $\vec{d}$ are two vectors satisfying $\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$ and $\vec{a} \cdot \vec{d} = 0$. Then find the vector $\vec{d}$ in terms of $\vec{a}, \vec{b}, \vec{c}$.
MVT248
11. Let $ABCD$ be a parallelogram such that $\overrightarrow{AB} = \vec{q}$, $\overrightarrow{AD} = \vec{p}$ and $\angle BAD$ be an acute angle. If $\vec{r}$ is the vector that coincides with the altitude directed from the vertex B to the side AD , then find $\vec{r}$ in terms of $\vec{p}$ and $\vec{q}$.
MVT249
12. If $\vec{a}, \vec{b}$ be two perpendicular unit vectors such that $\vec{x} = \vec{b} - (\vec{a} \times \vec{x})$, then find $|\vec{x}|$.
MVT250
13. Let $\vec{u} = \hat{i} + \hat{j}, \vec{v} = \hat{i} - \hat{j}$ and $\vec{w} = \hat{i} + 2\hat{j} + 3\hat{k}$. If $\vec{n}$ is a unit vector such that $\vec{u} \cdot \vec{n} = 0$ and $\vec{v} \cdot \vec{n} = 0$, then find $2|\vec{w} \cdot \vec{n}| + |\vec{u}|^2 + |\vec{v}|$.
MVT251
14. Given $|\vec{a}| = |\vec{b}| = 1$ and $|\vec{a} + \vec{b}| = \sqrt{3}$. If ' $\vec{c}$ ' is a vector such that $\vec{c} - \vec{a} - 2\vec{b} = 3(\vec{a} \times \vec{b})$, then find $\vec{c} \cdot \vec{b}$.
MVT252

ANSWER KEY

1. C 2. A 3. 3 7. 64 8. $-\hat{i} + \hat{k}$ 9. $\vec{b} = -\hat{i} + \hat{j} - 2\hat{k}$
10. $\vec{d} = \vec{c} - \left(\frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} \right) \vec{b}$ 11. $\left(\frac{\vec{q} \cdot \vec{p}}{|\vec{p}|} \right) \vec{p} - \vec{q}$ 12. $\frac{1}{\sqrt{2}}$ 13. $8 + \sqrt{2}$
14. $\frac{5}{2}$